

GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 4 of 6

The Design Thinking Process

Empathise, Define, Ideate, Prototype, Test · IDEO's three spaces · the Double Diamond · what every model of the process agrees on

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In this presentation

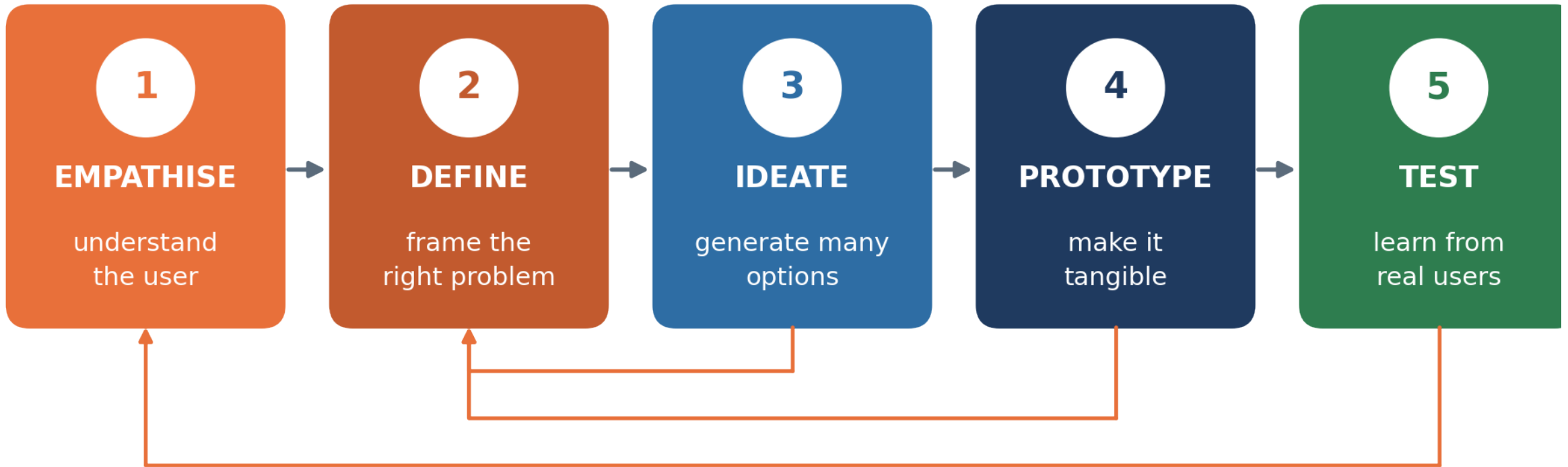
- The five-stage model
- Empathise and Define
- Ideate
- Prototype and Test
- IDEO's three spaces
- Three models compared
- What every model agrees on
- Common misunderstandings

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The five stages

What happens in each, and what each one produces

The five-stage model



the loops matter as much as the stages

The Stanford d.school model, which this course follows.

Stage 1 – Empathise

- Set aside what you assume and find out how the user actually experiences the problem
- Observe people in their own setting; interview them about specific past events
- Look for what people do, not only for what they say they do
- **Produces:** field notes, photographs, quotations, empathy maps

KEY POINT You cannot empathise from a classroom. This stage happens where the problem lives.

Stage 2 – Define

- Turn everything you gathered into one clear, actionable problem statement
- Name the user, their need, and the insight that explains the need
- A good statement is narrow enough to act on, wide enough to allow many answers
- **Produces:** a point-of-view statement and “How might we...” questions

KEY POINT This is the stage engineers most often skip – and it is the one that decides whether the rest is worth doing.

Stage 3 – Ideate

- Generate a wide field of options before choosing any one of them
- Use structured techniques – brainstorming, mind mapping, SCAMPER, lateral thinking
- Then converge deliberately, with criteria you agreed in advance
- **Produces:** a large set of ideas, and two or three chosen to build

KEY POINT The quality of your final idea depends on how many you were willing to have before choosing.

Stage 4 – Prototype

- Make the idea tangible as cheaply and as quickly as you can
- Paper, cardboard, clay, a role play, a slide sequence – fidelity stays low on purpose
- Each prototype should answer one specific question you are unsure about
- **Produces:** something a user can hold, use or walk through

KEY POINT A prototype is not a small version of the product. It is an experiment with a question attached.

Stage 5 – Test

- Put the prototype in front of real users and watch what they do with it
- Ask them to use it, not to judge it; silence is data too
- Record what surprised you – surprises are where the learning is
- **Produces:** evidence that sends you back to an earlier stage

KEY POINT Testing rarely ends a project. It usually sends you back to Define or Ideate with better information.

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Other maps of the same territory

Different models, the same underlying moves

IDEO's three spaces of innovation



overlapping spaces, not sequential steps

Tim Brown describes three overlapping spaces rather than a sequence of steps.

Three models compared

d.school – 5 stages	IDEO – 3 spaces	Design Council – 4 Ds
Empathise	Inspiration	Discover
Define	Inspiration	Define
Ideate	Ideation	Develop
Prototype	Ideation	Develop
Test	Implementation	Deliver

Different names, the same moves: understand, frame, generate, build, learn.

What every model agrees on

- Understand the people before you decide what to build
- Frame the problem explicitly, and in writing
- Open the solution space wide before you close it
- Learn by building something rough, not by arguing about it
- Expect to go round again – the loop is the method, not a sign of failure

KEY POINT If a model has no loop in it, it is not describing design.

Common misunderstandings

What people assume

- It runs once, front to back
- Empathy means being nice
- Ideation is one brainstorm
- A prototype must look finished

What actually happens

- It loops, often many times
- Empathy means gathering evidence
- Ideation is structured and timed
- A rough prototype teaches more

KEY POINT The stages are a checklist of moves to make, not a timetable to march through once.

THINK ABOUT IT

Your team has two weeks and no money. Which stage would you cut – and what would it cost you?

- What goes wrong in a project that skips Empathise?
- What goes wrong in one that skips Define?
- Can you prototype before you have ideated? When would that help?
- Which stage is hardest to do honestly, and why?

Summary

- Five stages: Empathise, Define, Ideate, Prototype, Test
- Each stage produces something specific that the next stage consumes
- Define is the stage engineers skip and the one that decides the outcome
- IDEO describes three overlapping spaces; the Design Council uses four Ds
- All three models make the same moves under different names
- Every model loops – a process without a loop is not design

Next: Human-centred design and the engineer's role

- What human-centred really means
- Desirability, feasibility and viability
- Users, customers and stakeholders
- Engineering as a societal activity, and its ethics